

Software Requirements – Blockchain-based Ayurvedic Herb Traceability System

This document details the complete software stack required to implement the blockchain-based Ayurvedic herb traceability proof-of-concept, covering blockchain, backend services, mobile/web interfaces, and integration tools.

Sr. No.	Software / Tool Name	Cost	Usage Layer	Description & Official Links
1	Hyperledger Fabric	Free	Blockchain	Permissioned blockchain framework for supply-chain traceability. Link: https://www.hyperledger.org/use/fabric
2	Docker & Docker Compose	Free	Infrastructure	Containerization of blockchain nodes and services. Link: https://www.docker.com/
3	Node.js	Free	Backend	REST APIs and chaincode support. Link: https://nodejs.org/
4	MongoDB	Free	Database	Off-chain storage for dashboards and analytics. Link: https://www.mongodb.com/
5	React.js	Free	Frontend	Consumer portal and admin dashboard UI. Link: https://react.dev/
6	Leaflet.js	Free	Maps	Interactive geo-location maps for provenance display. Link: https://leafletjs.com/
7	QR Code Generator Library	Free	Utility	Generates batch-level QR codes. Link: https://www.npmjs.com/package/qrcode
8	Ubuntu Server OS	Free	Operating System	Server OS for hosting blockchain nodes. Link: https://ubuntu.com/